Evolving a Set of Competing Economic-School Prompts: NSGA-II with LLM-Guided Mutation

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Abstract

LLM agents that speak from different perspectives are only useful if each agent stays true to its own perspective and the agents do not all sound the same. Prompt-tuning research usually improves one prompt for one score, so it does not handle the trade-off between three goals directly: staying faithful to the school, being different from the other schools, and keeping the prompt short. We treat the three school prompts as one set. We search for good sets with NSGA-II, a standard multi-objective evolutionary algorithm, and add a new kind of mutation: an LLM rewrites one part of a prompt and is told which goal to improve. In the main experiment (model `qwen3.8-max`, population 6, four generations, three seeds), the choice of selection method made no visible difference. The guidance made the difference. Guided LLM edits reach a hypervolume (a single score for how much of the trade-off space a set covers; higher is better) of about 0.12, compared with about 0.10 for rule-based edits and for LLM rewrites with no goal. The guided edits win in every seed. We also repeated every arm with Claude Haiku 5.5 and five seeds. The result points the same way in four of five seeds, but less clearly, because a length limit tuned for the first model rejected half of the Haiku edits. Compared with the published EvoPrompt-GA method, the original version fails here because its prompts grow too long for the agents to answer. A length-limited version, added after that result, still loses to our method in all five seeds (one-sided $p = 0.031$). On held-out events, no method beats the hand-written prompts after correction for multiple comparisons, on either model. Random search gains about as much fidelity as evolution does. The searched prompt sets are 20–30% shorter, with similar fidelity and disagreement. All protocols, raw model outputs (13,121 calls) and analysis code are released.

Keywords: evolutionary prompt optimisation, multi-objective evolutionary algorithms, NSGA-II, LLM-guided mutation, multi-agent diversity, quality-diversity.

1 Introduction

1.1 Problem and motivation

Systems that combine several LLM agents, each speaking from a different perspective, are increasingly used for analysis, forecasting and debate (Baltaji, Hemmatian, and Varshney 2024; Takano, Hirano, and Nakagawa 2025). Their value rests on two properties that pull in opposite

directions. Each agent must stay *faithful* to its perspective, and the agents must be *distinct* from one another, or the ensemble collapses into one voice. Recent evidence points to exactly this failure: debate and ensemble systems lose diversity through conformity and majority pressure (H. Wu, Li, and Li 2025; Zhu et al. 2026), and aligned models produce homogeneous outputs across many open-ended queries (Jiang et al. 2025). Prompts are the cheapest lever for setting perspective, but they are usually written by hand.

Evolutionary prompt optimisation can search prompts automatically, with an LLM supplying the variation operators (Fernando et al. 2024; Guo et al. 2024; Lange, Tian, and Tang 2024). These methods, however, optimise one prompt against one scalar score. None of them evolves a *set* of prompts whose members must be good individually and different from each other, and none exposes the trade-off between fidelity, distinctness and cost (prompt length). Multi-objective prompt search has started to appear (Zhao et al. 2025; Câmara et al. 2025). We found no such work for multi-agent sets.

A preliminary, unpublished result from the author's MPhil project, whose harness this project borrows for its measurements, points the same way. In a 240-request disagreement matrix over six economic frameworks (synthetic events, generator seed 11, the same model identifier `qwen3.8-max`), the three trained schools disagreed about twice as much as one school disagreed with itself, while New Keynesian duplicated Keynesian within the noise floor. This evidence is used here only as motivation; none of its events or outputs enter the experiments below.

1.2 Research questions

- **RQ1.** Can NSGA-II over prompt sets, with LLM-guided mutation, find prompt sets that dominate a hand-written baseline and single-objective or random search in fidelity, inter-agent disagreement and length?
- **RQ2.** Which part matters most: the selection method, the LLM mutation itself, or the goal-directed guidance given to the LLM?
- **RQ3.** Do the gains hold on unseen events?
- **RQ4.** How does the method compare with a published evolutionary prompt optimiser (EvoPrompt-GA), and do the findings hold on a second model from another provider?

1.3 Contributions

1. **Representation and objectives.** An individual is a set of three school genes (diagnosis, concepts, stance, contrast) decoded through the unchanged thesis prompt scaffold. Objectives are school fidelity, inter-school disagreement and prompt length, with Deb's constraint handling on output validity.
2. **Operators.** Structural crossover at school and field level; rule mutation that touches one school per child; and an LLM-guided mutation in which the parent's weakest objective is turned into a rewrite instruction (AI-to-EC).
3. **Evaluation design.** We compare the two selection methods and the two mutation types in a 2 × 2 ablation. We add a control in which the LLM rewrites prompts without being told a goal, a comparison with a published method (EvoPrompt-GA), and a repeat of every arm with a second model (Claude Haiku 5.5, five seeds).

4. **Protocol.** We wrote the analysis plan down before the analysis, plus four later addenda. Each addendum was committed before the runs it covers (Section 4.8). Held-out events come from a separate seed. Significance tests are exact sign-flip tests with Holm correction. Every model output is cached, so the analysis can be rerun offline from the committed files.

2 Related Work

We collected 34 papers from 2024–2026 (`report/references.bib`). We checked every author list and first-submission date against arXiv, and the journal details against Crossref. The papers are grouped by theme, not summarised one by one. Promptbreeder, EvoPrompt and Eureka are included even though their first arXiv versions are from 2023, because their conference versions are from 2024.

2.1 LLMs as evolutionary variation operators

The strongest current trend treats an LLM as the mutation and crossover operator of an evolutionary loop. Promptbreeder evolves task prompts and the mutation prompts themselves (Fernando et al. 2024); EvoPrompt connects LLM-generated crossover and mutation with GA and DE on discrete prompts (Guo et al. 2024); EvoLLM treats an LLM as an in-context recombination step for evolution strategies (Lange, Tian, and Tang 2024). Beyond prompts, LLMs design heuristics and algorithms: EoH co-evolves natural-language thoughts with code (F. Liu et al. 2024), ReEvo adds reflective, verbal feedback to crossover and mutation (Ye et al. 2024), LLaMEA generates metaheuristics that beat CMA-ES on BBOB (Stein and Bäck 2024), and Eureka evolves reward code from simulator feedback (Ma et al. 2024). Automated design of agentic systems applies the same idea to whole agents, with an archive rather than a selection population (S. Hu, Lu, and Clune 2025). Surveys place these methods in a common frame, with the LLM as optimiser, predictor, extractor or designer (X. Wu et al. 2025; F. Liu et al. 2026). Recent work also adds guidance on which parent and which operator to call, to save model calls (Wang et al. 2026).

Gap. Nearly all of this work optimises a single fitness value. Where diversity appears, it is an exploration device inside one search, not an objective.

2.2 Multi-objective and quality-diversity search with LLMs

Pareto-based prompt search is emerging. Pareto Prompt Optimization trains a prompt generator with dominance-based preference losses instead of a fixed scalarisation (Zhao et al. 2025). MOPrompt runs a multi-objective evolutionary search on accuracy versus context length and reports a 31% shorter prompt at the best baseline's accuracy on one sentiment task (Câmara et al. 2025). REMoH wraps NSGA-II around LLM-generated heuristics, with a clustering reflection step that keeps offspring diverse (Forniés-Tabuenca et al. 2025); an LLM-aided MOEA uses an LLM only when a surrogate says it is worth the call (W. Liu, Chen, and Tang 2024). On the quality-diversity side, LLMs can propose archive members in context (Lim, Flageat, and Cully 2024), MAP-Elites explores prompt space by structural descriptors (Santos, Julia, and Nascimento 2025), and CycleQD cycles which objective acts as the quality measure while the others act as behaviour axes (Kuroki et al. 2025).

Gap. Multi-objective LLM search so far targets accuracy versus cost on one task. None uses a *between-agent* distinctness objective, and none evaluates a prompt set on held-out events.

2.3 Diversity and perspective fidelity in multi-agent LLM systems

Multi-agent discussion can increase viewpoint diversity, but conformity and confabulation erode it. Persona-assigned agents drift from their assigned views under debate instructions (Baltaji, Hemmatian, and Varshney 2024). Vanilla debate often underperforms majority vote, and explicitly diversifying the initial viewpoints and calibrating confidence helps (Zhu et al. 2026). In a controlled logic study, group diversity and reasoning strength mattered most, while majority pressure discouraged independent correction (H. Wu, Li, and Li 2025). Diversity collapses across models, not only within one: an open-ended study finds intra-model repetition and inter-model homogeneity (Jiang et al. 2025). For economic reasoning, LLMs lean towards intervention-oriented causal conclusions, and that bias persists under one-shot prompting (Lee et al. 2026). Macro agents with explicit hawkish and dovish beliefs can improve FOMC classification, but the study does not check whether agents stay faithful to their belief across rounds (Takano, Hirano, and Nakagawa 2025). Survey simulations with LLM agents match macro expectation distributions only when priors are injected (Lin, Sun, and Yan 2025).

Automated multi-agent design is also active. MASS optimises prompts and topology in three stages with a single task score (Zhou et al. 2026). EvoMAS evolves multi-agent configurations by mutation and crossover guided by execution traces (Y. Hu et al. 2026).

Gap. Diversity is studied as an accuracy device or as a property to be measured, but not as an optimisation target next to persona fidelity, and not for competing economic perspectives.

2.4 Evolutionary model merging, fine-tuning and self-evolution

This family is on the course's list of directions and is the closest population-based alternative to our method that changes model weights instead of prompt text. We review it briefly to show where our setting sits.

Evolution has been applied to parameters and recipes rather than text. Evolutionary search over merge recipes, in parameter and data-flow space, yields state-of-the-art Japanese models (Akiba et al. 2025). Multi-objective evolution of merged experts trades cross-task performance against storage (H. Qiu et al. 2026), and a two-dimensional recombination of LLM blocks is searched by bi-level multi-objective evolution (X. Wu, Zhou, and Tan 2026). Layer-wise merging of reasoning-tuned LLMs is another route to combining specialised models (Zhang et al. 2024). Evolution strategies can fine-tune billion-parameter LLMs without gradients (X. Qiu et al. 2026), and black-box evolutionary merging works from API queries alone (Chen et al. 2025). Self-evolving training co-evolves a rubric generator and a policy without human labels (Li et al. 2026).

Gap. These methods optimise weights, which requires training or merging infrastructure. The project's setting, where the only decision variables are the texts of a few prompts, is a cheaper and more interpretable point in the same family.

2.5 Synthesis and positioning

Table 1 summarises the literature along the axes that matter for this project.

Axis	Typical finding	Relevance to our design
LLM as operator	Works for prompts, heuristics and code (Fernando et al. 2024; Guo et al. 2024; F. Liu et al. 2024)	We use an LLM mutation with an objective-directed instruction
Objective	Mostly one scalar; Pareto variants exist for accuracy versus cost (Zhao et al. 2025; Câmara et al. 2025)	Three objectives, one of which is inter-agent distinctness
Diversity	Measured or used as initialisation; homogeneity documented (Jiang et al. 2025; Zhu et al. 2026)	Made an explicit objective
Persona fidelity	Erodes under debate (Baltaji, Hemmatian, and Varshney 2024; H. Wu, Li, and Li 2025)	Measured with a lexicon; held out on separate events
Evaluation	Mostly one task; few held-out checks	Search and held-out seeds are separate; ablations are reported

Our position. To our knowledge no paper combines (i) a *set* of agent prompts as one individual, (ii) a between-agent distinctness objective next to fidelity and length, (iii) an LLM as the mutation operator guided by the weakest objective, and (iv) a held-out check on unseen events. We treat the position as a gap identified from this review rather than as a proof of novelty.

3 Method

3.1 Problem formulation

An individual is a prompt set $P = (g_{\text{kw}}, g_{\text{au}}, g_{\text{mo}})$ with one gene per school. For a set of synthetic events E, each school s answers each event with a parseable report or fails. Three objectives are minimised, all in [0,1]:

- $g_1 = 1 - \text{fidelity}(P)$, where fidelity is the mean lexicon consistency of each school's answers (the thesis consistency score, C_s).
- $g_2 = 1 - \text{disagreement}(P)$, where disagreement is the mean pairwise signal disagreement between the three schools (`mosaic.divergence.distance`).
- $g_3 = \min(1, \text{words}(P)/600)$.

Feasibility requires that at least 90% of school outputs are parseable. Infeasible sets are ranked by their violation, following Deb's rule.

3.2 Genome and decoding

Each school gene has four parts: `diagnosis` (sentences), `concepts` (an ordered subset of a concept pool), `stance` (sentences describing typical readings of events) and an optional `contrast` (sentences of the form "unlike the other schools, you …"). Decoding fills the unchanged

`mosaic.divergence.messages()` template, so the output format (causal chain, scenarios, signals) and the events are identical to the thesis harness. The initial population is the hand-written set B0 plus perturbed copies of it.

3.3 Variation

- **Structural crossover** (probability 0.7). With probability 0.5 whole school genes are swapped between parents; otherwise each field of each school is picked from one parent.
- **Rule mutation.** Pick one school, then apply one or two small edits: remove or swap a sentence of its diagnosis, or remove, add or reorder a concept. Only one school changes, so the other two schools' outputs can come from the cache. None of these edits adds a diagnosis sentence or changes `stance` or `contrast`. So rule mutation can only keep or shorten the text it edits (adding a concept adds only a few words). This pushes towards shorter sets, which matters when reading the length results (Section 5.2).
- **LLM-guided mutation** (probability 0.5 of the mutations in the LLM methods). The parent's evaluation picks the objective to improve, with probability proportional to its badness: fidelity (rewrite the least faithful school's diagnosis using its lexicon terms), distinctness (rewrite one field of one of the two most similar schools so that it makes different calls from the other), or length (shorten the longest school's diagnosis to 40 words). The model returns one paragraph; it is accepted if it has 5 to 120 words, and otherwise the child falls back to rule mutation (on qwen, summed over the three seeds, 8 of 30 LLM children in NSGA-II and 10 of 34 in the GA were rejected this way; on Haiku about half, Section 5.3).

3.4 Selection

- **NSGA-II.** Constrained non-dominated sorting, crowding distance, binary tournament on (rank, crowding), and elitist (μ+λ) environmental selection (Deb et al. 2002).
- **GA (B2).** The same operators, but selection by feasibility first and then the equal-weight sum of g_1, g_2, g_3. This isolates the effect of the selection scheme.

3.5 Baselines, ablation and indicator

- **B0.** The hand-written profiles, evaluated with three independent sampling seeds.
- **B1 random search.** Independent random rule perturbations of B0 (1 to 3 operators per school), the same number of evaluations, and the same archive.
- **Ablations.** NSGA-II with rule mutation only, and the GA with rule mutation only; with the two LLM arms they form a 2 × 2 design (selection × mutation).
- **Unguided LLM control.** NSGA-II with an LLM mutation that carries no objective information: the same call, share and acceptance rule as the guided operator, but the school and the field (diagnosis or stance) are drawn at random and the instruction is fixed: "Rewrite this paragraph in your own words. Keep its meaning and keep it as one paragraph." It separates the guidance from LLM rewriting as such.
- **Published baseline: EvoPrompt (GA)** (Guo et al. 2024). Parents are chosen by roulette wheel on the score $1 - \bar{g}$; one LLM call per school and child applies the paper's crossover-and-mutation template (verbatim, Figure 1 of the paper) to the two parents' diagnosis paragraphs, and the best N of parents and children survive. Four adaptations were needed and are stated as

such: the operator is applied to each of the three schools (the paper evolves one prompt); the answer inside `<prompt>` brackets is accepted if it has 5 to 600 words (the paper's operator writes whole prompts, see Section 4.8); the final solution is the shared knee, not EvoPrompt's argmax; and the initial population is the shared one (B0 plus perturbed copies), as the paper also starts from manual prompts.

- **Indicator.** The hypervolume: a single number for how much of the trade-off space a set of solutions covers (higher is better). It is computed exactly in three dimensions, with reference point (1, 1, 1). The archive keeps every feasible set that was evaluated, so all methods are compared on the same footing.
- **Knee.** The archive member closest to the ideal point (0, 0, 0), that is, the best overall compromise. We wrote this rule down before the held-out analysis (`code/PROTOCOL.md`). We do not rescale the objectives. Across the final fronts, the length score ranges from about 0.40 to 0.79 and the disagreement score from about 0.50 to 0.78, but the fidelity score only from about 0.47 to 0.60. So fidelity counts for least in the distance, and the knee tends to be one of the shorter feasible sets.

4 Experimental Setup

4.1 Task and data

The task is a set of synthetic macroeconomic events produced by `mosaic.teacher.make_items`. The search set is the first five events of seed 23 and the held-out set is the first ten of seed 29; one held-out event (`syn-k0007`, seed 29) has the same text as one search event (`syn-k0002`, seed 23), although the two have different macro contexts. Section 5.2 reports the held-out tests with that event removed. Seed 11 (the thesis disagreement matrix) and all thesis held-out files are not used. Lexicons come from `configs/lexicons.toml`, which the thesis marks as PROVISIONAL; this affects the fidelity objective (Section 6).

4.2 Model and backend

Main model. Each call goes to an OpenAI-compatible API (`qwen3.8-max`, the model used in the thesis disagreement matrix) with temperature 0.7, top-p 0.9, at most 1,024 output tokens, thinking turned off, and a sampling seed for each run. Every response is saved to a cache keyed by the model, the messages and the sampling settings (`results/cache/llm_cache.jsonl`). The qwen calls used a prepaid subscription with no per-call price. The cost figures are therefore estimates: we record the measured token counts and multiply them by assumed list prices (USD 2.4 and 9.6 per million input and output tokens). The estimate is only a budget check. The token counts let anyone redo it with a real price list.

Second model. The replication uses `claude-haiku-5-5` through the official Anthropic SDK. The model rejects non-default temperature and top-p, so its default sampling is used; thinking is disabled and the output limit is 1,024 tokens, as for qwen. The API has no sampling seed, so the seed is kept in the cache key (each seed draws its own samples) but not sent. Everything for this model is stored under `results/haiku/` with its own cache; its ledger uses the list prices of USD 0.10 and 0.50 per million tokens.

4.3 Hyperparameters

Parameter	Value
Population N	6
Generations G	4 (so $N(G+1) = 30$ evaluations per run)
Search events / held-out events	5 / 10
Algorithm seeds (paired across methods)	1, 2, 3 (qwen); 1 to 5 (Haiku)
Crossover / mutation / LLM share	0.7 / 0.9 / 0.5
Tournament size	2
Feasibility threshold	0.9
Length normalisation	600 words
Hypervolume reference	(1, 1, 1)
LLM edit length	5 to 120 words (guided and unguided edits); 5 to 600 words (EvoPrompt answers)
Sampling	qwen: temperature 0.7, top-p 0.9; Haiku: provider default; both max 1,024 tokens, thinking off

4.4 Procedure

1. For each method and seed, run the search on the five search events and keep the archive (`results/runs/<method>_s<seed>/gen_log.csv`, `evaluations.jsonl`, `front.jsonl`).
2. Select the knee of the final archive and write it to `results/best_prompts/`.
3. Re-measure each knee, and B0, on the ten held-out events. Each knee is also re-scored from the cache per event (`results/heldout_per_event.csv`).
4. Average over seeds per event and test each method against B0 with an exact paired sign-flip test over the ten events, with Holm over the method-versus-B0 tests of that model (`results/heldout_tests.csv`); the protocol family adds every method-versus-method test (`results/heldout_tests_protocol.csv`).

4.5 Statistics

Search-level results are mean ± sample standard deviation over three seeds. Held-out intervals are 95% bootstrap intervals over the ten events (10,000 resamples, `mosaic.stats.paired`, seed 0). These intervals are not corrected for multiple comparisons. The p-values come from an exact sign-flip test and are the ones corrected with Holm. With three paired seeds the smallest possible one-sided p-value is 1/8, so the qwen search-level comparisons are descriptive only; with five seeds (Haiku) it is 1/32. The held-out test uses the ten events, where the smallest possible two-sided p-value is 2/1024.

4.6 Reproduction

`code/README.md` gives the commands. The qwen matrix was run with `python -m evo_prompt.runner --methods b0 nsga2 ga random nsga2_rule --seeds 1 2 3 --gens 4 --pop 6 --heldout --budget 30`, the two ablation arms with `--methods ga_rule nsga2_llmu`

and the same settings, and the Haiku replication with `--profile haiku` and seeds 1 to 5 (the exact commands are in the protocol addenda). The tables come from `python -m evo_prompt.analyze` (add `--profile haiku` for the replication) and the figures from `python -m evo_prompt.figures`. The analysis reads only the committed cache, so it makes no model calls.

4.7 Computing environment and measured performance

Hardware. The search and the analysis ran on a CPU-only Linux machine (x86_64, no GPU). The language model was accessed over HTTPS through an OpenAI-compatible chat endpoint, so the model's hardware is outside our control.

Software. Python 3.13.18; numpy 2.5.3 (analysis), matplotlib 3.11.2 (figures), pytest 9.2.0 (tests). The qwen client uses only the Python standard library; the Haiku client uses the official `anthropic` SDK 2.3.0. The versions are those of the environment in which the reported runs were made; no lock file is provided.

Concurrency. On qwen each batch of model calls sends up to 16 requests in parallel; on Haiku up to 8, and from the parallel phase on (Section 4.8) up to 6 per process with up to 7 processes. Every response is written to the cache before the next batch starts.

Measured performance. The figures below come from the caches, which store the response time and token counts of every call.

- qwen, recorded calls: 3,709, of which 2,743 belong to the main matrix, 106 to the API smoke test, 675 to the two ablation arms and 185 to the incomplete EvoPrompt run.
- qwen, per-call latency: median 14.9 s, mean 15.0 s, 90th percentile 17.3 s, maximum 183.9 s; token totals 2,510,128 input and 2,217,418 output.
- Haiku, recorded calls: 9,412 (the smoke test of addendum B is not included); per-call latency median 5.6 s, mean 5.7 s, 90th percentile 7.0 s, maximum 10.0 s; token totals 9,261,362 input and 8,710,072 output; summed latency about 14.9 hours.
- Summing the qwen per-call latencies gives about 15.4 hours. This is not wall-clock time, because requests ran in parallel. Wall-clock durations were not logged.
- The 27 offline tests run in about one second.

4.8 Protocol, addenda and deviations

The analysis follows `code/PROTOCOL.md` and four addenda. Each addendum was committed before the first model call of the runs it governs; the commit times (UTC, 2026-11-02) are in the git history.

Document	Commit	Governs
`PROTOCOL.md`	`b3cfd5e` (2026-11-01)	main qwen matrix (B0, NSGA-II + LLM, GA, random search, NSGA-II rule only)
Addendum A	`78cb5a1` (04:19)	GA rule only and NSGA-II + unguided LLM on qwen; questions Q1 to Q3 and their decision rule

Document	Commit	Governs
Addendum B	`c58798a` (04:29)	Haiku replication of every arm, seeds 1 to 5
Addendum C	`507545a` (04:37)	EvoPrompt-GA baseline; parallel execution of the Haiku runs
Addendum D	`f5e935e` (05:00)	length-bounded EvoPrompt variant on Haiku (post hoc, see item 7)

The analysis departs from these documents, or from their wording, in the following places.

1. **Timing of the main protocol.** `PROTOCOL.md` says it was written before any full-matrix result existed. In fact it was committed on 2026-11-01 at 15:12 UTC, while the matrix was still running. That commit already included the seed-1 knee prompt sets of NSGA-II, the GA and random search. The runner measures each knee on the held-out events at the end of its run, so some seed-1 held-out values may already have existed. The held-out tables, tests and figures were produced afterwards. So the main protocol was fixed before the held-out analysis, not before every result. We do not call it a pre-registration. The addenda do not have this problem.
2. **Test family.** The protocol applies Holm across method-versus-B0 *and* method-versus-method comparisons. Tables 3 and 4 use the method-versus-B0 tests only (12 tests for qwen). The full family (12 + 30 = 42 tests for qwen, 16 + 56 = 72 for Haiku) is in `results/heldout_tests_protocol.csv` and its Haiku counterpart; the smallest Holm-adjusted p-value is 1.0 for qwen and 0.425 for Haiku, so no conclusion changes.
3. **Post-hoc sensitivity analysis.** Removing the held-out event whose text duplicates a search event (Section 5.2) was not in the protocol. It was added after the overlap was found.
4. **EvoPrompt acceptance rule (amendment to addendum C).** Addendum C first applied the 5-to-120-word rule of our edit operator to EvoPrompt's answers. All 18 first-generation answers on qwen had 143 to 370 words, so every one was rejected and the baseline would have run without variation. Before any fitness comparison, the rule was changed for this arm to 5 to 600 words (`187dc01`, 04:38); long answers are penalised by the length objective instead of being discarded. The stopped first run had already written its seed-1 search files; they were moved out of `results/` without being read.
5. **qwen EvoPrompt incomplete.** The qwen quota ran out during generation 2 of seed 1 of the restarted EvoPrompt run. As addendum C requires, the arm is reported as incomplete and left out of every qwen table; the comparison with the published baseline is made on Haiku only.
6. **Parallel execution on Haiku.** For speed, the sequential Haiku run was stopped after B0 and NSGA-II seeds 1 to 3, its cache was frozen as a read-only base, and the remaining arms (later also individual seeds of the two slowest arms) ran in separate processes with their own cache layers. Arms therefore do not share samples drawn after the freeze, so the Haiku trajectories of, for example, NSGA-II and the GA are less coupled than on qwen. This changes only execution, not the design. It also means that the initial populations of seeds 4 and 5 were sampled separately per arm, so generation-0 hypervolumes differ slightly between Haiku arms (Figure 4).
7. **Post-hoc variant.** Addendum D was written after the Haiku EvoPrompt result had been seen and analysed. It adds the smallest change that lets that baseline work in this problem; the

verbatim baseline is reported as well, and the variant is labelled as post hoc wherever it appears.

5 Results and Analysis

5.1 Main model: search level

Table 2 reports the archive and the knee on the search events for the six qwen arms. Values are mean ± sd over three seeds (`results/summary_table.md`).

Method	Hypervolume	Front size	Knee fidelity	Knee disagreement	Knee validity	Knee words
NSGA-II + LLM	0.1211 ± 0.0182	9.3 ± 1.5	0.461 ± 0.026	0.460 ± 0.025	0.978 ± 0.039	330 ± 50
GA (weighted sum)	0.1220 ± 0.0130	10.7 ± 1.2	0.452 ± 0.038	0.459 ± 0.032	1.000 ± 0.000	305 ± 29
Random search	0.0978 ± 0.0119	4.3 ± 0.6	0.474 ± 0.011	0.437 ± 0.017	1.000 ± 0.000	350 ± 38
NSGA-II, rule only	0.0979 ± 0.0135	7.7 ± 4.0	0.460 ± 0.016	0.429 ± 0.061	0.933 ± 0.000	346 ± 37
GA, rule only	0.0985 ± 0.0201	7.7 ± 2.1	0.456 ± 0.032	0.463 ± 0.026	0.933 ± 0.000	356 ± 52
NSGA-II + unguided LLM	0.1022 ± 0.0320	9.7 ± 2.9	0.459 ± 0.027	0.418 ± 0.013	1.000 ± 0.000	338 ± 71

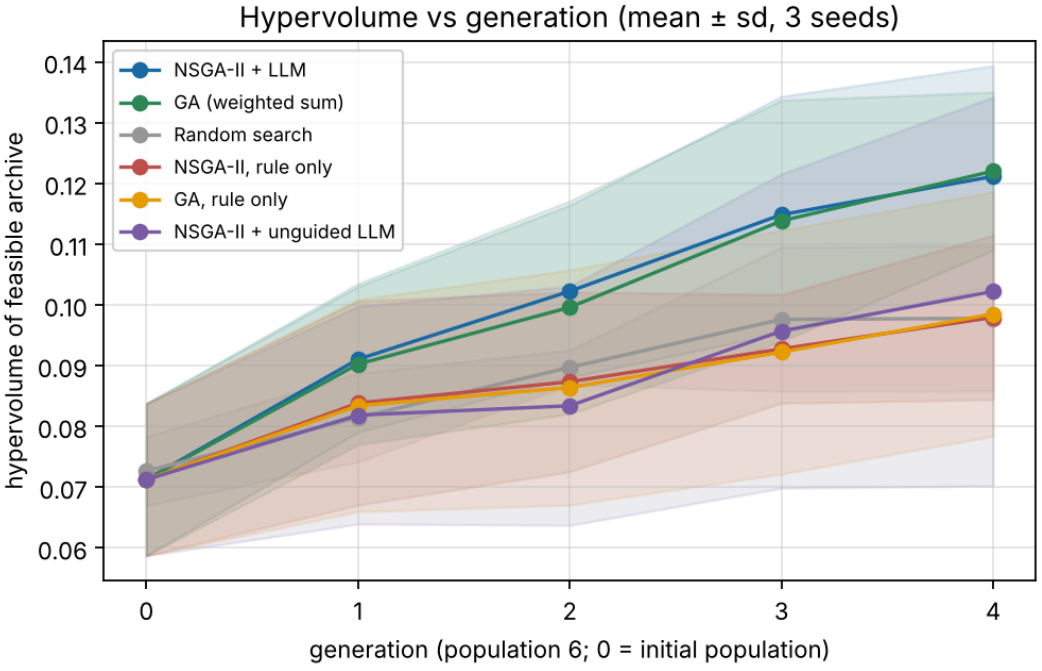


Figure 1. Mean hypervolume of the feasible archive per generation on qwen (3 seeds, shaded ±1 sd). Generation 0 is the shared initial population.

Figure 1 shows the mean hypervolume per generation. Generation 0 is the shared initial population (0.071 for every arm except random search, 0.073). The two arms with guided LLM mutation rise above the others from generation 1 on and end at 0.121 and 0.122; the three rule-based arms and the unguided control end between 0.098 and 0.102.

Table 2b arranges the final hypervolume as the 2 × 2 design of selection and mutation, with the unguided control alongside.

Selection	Rule mutation only	Guided LLM mutation	Unguided LLM mutation
GA (weighted sum)	0.0985	0.1220	not run
NSGA-II	0.0979	0.1211	0.1022

Addendum A fixed three questions and a decision rule before the two new arms were run: a direction counts as consistent only if all three paired seeds agree in sign (one-sided sign-flip p = 1/8, the smallest value three seeds allow).

Question	Comparison	Seed 1	Seed 2	Seed 3	Mean	Consistent
Q1 selection, rule mutation	GA rule − NSGA-II rule	−0.0019	−0.0049	+0.0085	+0.0006	no
Q2 LLM mutation, GA selection	GA − GA rule	+0.0235	+0.0463	+0.0007	+0.0235	yes
Q3 guidance	NSGA-II + LLM − NSGA-II + unguided LLM	+0.0327	+0.0212	+0.0029	+0.0189	yes

The answer to RQ2 on qwen is now direct.

- **Selection does not matter here.** With rule mutation, the two selection methods reach the same hypervolume, and their differences change sign across seeds (Q1). With guided LLM mutation they also end level (0.1211 against 0.1220). Four generations with population 6 give the methods little room to differ, so this claim is about this budget only.
- **LLM mutation helps under both selection methods.** With GA selection it beats rule mutation in all three seeds (Q2). With NSGA-II the same holds against the rule-only version (0.114 against 0.087, 0.108 against 0.093, and 0.142 against 0.113).
- **The guidance is what helps, not LLM rewriting in general.** The unguided control ends at 0.102, close to the rule-only versions. The guided operator beats it in all three seeds (Q3). This is the most direct evidence for the operator proposed in Section 3.3.
- **How strong is this?** All three claims rest on three paired seeds. Some margins are small (for example, +0.0007 and +0.0029 for Q2 and Q3 at seed 3), and random search with seed 2 (0.112) ends above NSGA-II with LLM mutation at seed 2 (0.108). Section 5.3 repeats the questions with a second model and five seeds.

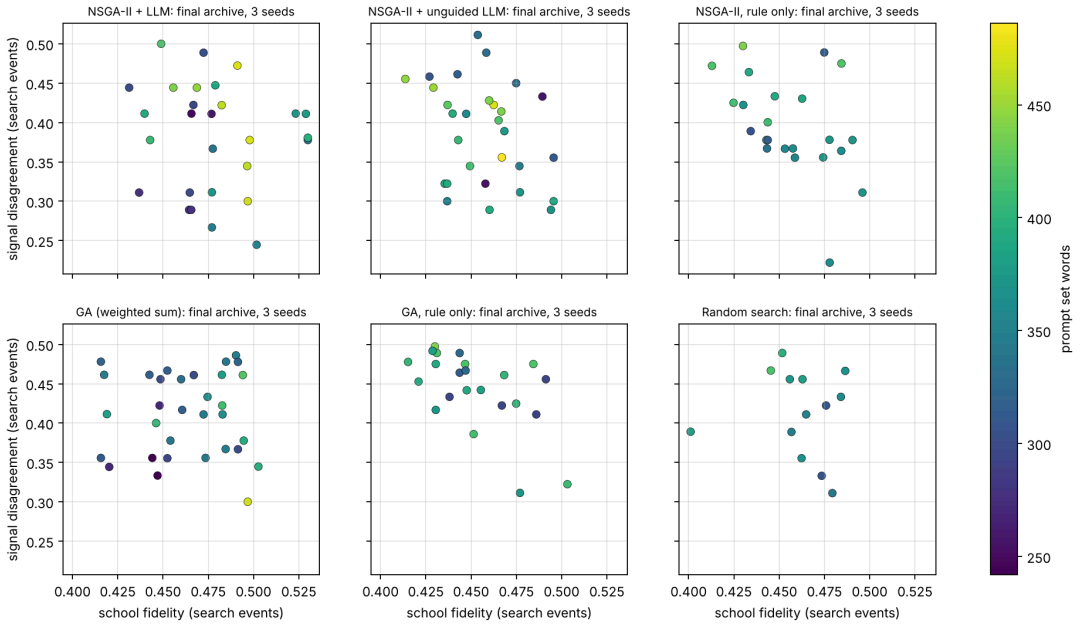


Figure 2. Final feasible archives of the qwen search arms (3 seeds each, pooled), fidelity against disagreement on the search events, coloured by prompt-set length on a common scale.

Figure 2 shows where the extra hypervolume comes from. The arms with LLM mutation reach the shortest sets (242 to 256 words); every rule-based arm stays at 292 words or more. The guided NSGA-II archive also reaches the highest fidelity (0.53), while the unguided control does not exceed 0.496.

5.2 Main model: held-out events

Table 3 reports the knee solutions on the ten held-out events. The ± values are not of the same kind: for B0 they are the spread over three sampling seeds of one fixed prompt set, and for the search methods they are the spread over three algorithm seeds, each of which produces a different knee prompt set (and is evaluated with its own sampling seed). The second includes the first.

Method	Fidelity	Disagreement	Validity	Words
B0 hand-written	0.406 ± 0.002	0.407 ± 0.028	0.989 ± 0.019	438
NSGA-II + LLM	0.424 ± 0.017	0.423 ± 0.006	0.989 ± 0.019	330 ± 50
GA (weighted sum)	0.448 ± 0.043	0.419 ± 0.027	0.978 ± 0.019	305 ± 29
Random search	0.432 ± 0.009	0.432 ± 0.024	0.967 ± 0.033	350 ± 38
NSGA-II, rule only	0.433 ± 0.032	0.428 ± 0.048	1.000 ± 0.000	346 ± 37
GA, rule only	0.433 ± 0.006	0.415 ± 0.075	1.000 ± 0.000	356 ± 52
NSGA-II + unguided LLM	0.441 ± 0.017	0.372 ± 0.037	1.000 ± 0.000	338 ± 71

Table 4 gives the paired tests. Each comparison uses the ten held-out events, and Holm correction covers the twelve method-versus-B0 tests.

Comparison	Metric	Mean difference (method − B0)	95% bootstrap CI	p (exact, raw)	p (Holm)
NSGA-II + LLM − B0	fidelity	+0.0179	[-0.0086, +0.0413]	0.2168	1.0000
NSGA-II + LLM − B0	disagreement	+0.0056	[-0.0649, +0.0725]	0.8672	1.0000
GA (weighted sum) − B0	fidelity	+0.0426	[+0.0133, +0.0734]	0.0293	0.3223
GA (weighted sum) − B0	disagreement	-0.0074	[-0.0889, +0.0704]	0.8848	1.0000
Random search − B0	fidelity	+0.0262	[+0.0084, +0.0437]	0.0254	0.3047
Random search − B0	disagreement	+0.0000	[-0.0813, +0.0741]	0.9805	1.0000
NSGA-II, rule only − B0	fidelity	+0.0272	[+0.0041, +0.0503]	0.0566	0.4531
NSGA-II, rule only − B0	disagreement	+0.0204	[-0.0518, +0.0852]	0.5664	1.0000
GA, rule only − B0	fidelity	+0.0275	[+0.0053, +0.0475]	0.0410	0.4102
GA, rule only − B0	disagreement	+0.0075	[-0.0610, +0.0704]	0.8086	1.0000
NSGA-II + unguided LLM − B0	fidelity	+0.0354	[+0.0080, +0.0632]	0.0488	0.4395
NSGA-II + unguided LLM − B0	disagreement	-0.0351	[-0.0907, +0.0278]	0.3242	1.0000

No comparison is significant after Holm correction (smallest adjusted p = 0.305). Before correction, the exact sign-flip p-values of the fidelity gains of the GA, random search, the GA with rule mutation and the unguided control lie between 0.025 and 0.049, and their uncorrected bootstrap intervals exclude zero; none survives the correction across twelve tests, and none survives in the 42-test protocol family (Section 4.8). We therefore read these fidelity gains as suggestive, not established. Removing the one held-out event whose text duplicates a search event (Section 4.1), the exact sign-flip p-values on the remaining nine events are 0.059 for the GA fidelity gain, 0.051 for random search, 0.109 for NSGA-II with rule mutation, 0.066 for the GA with rule mutation and 0.082 for the unguided control, and no conclusion changes.

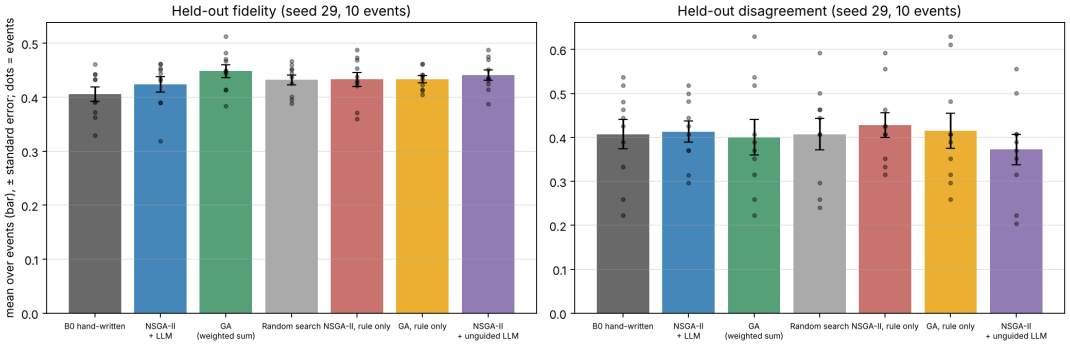


Figure 3. Held-out fidelity and disagreement of B0 and the qwen knee solutions. Bars are the mean over the ten events; error bars are standard errors; dots are individual events.

Four observations follow from the tables.

1. **The prompt sets get shorter, not less faithful.** The searched knees have 305 to 356 words against 438 for B0 (about 20% to 30% shorter), with fidelity and disagreement within a few hundredths of B0. Much of this is built into the design rather than found by the search: rule mutation can only shorten the diagnosis (Section 3.3), and the knee rule weights length more than fidelity (Section 3.5). Random search, which uses only rule mutation and the same knee rule, also ends 20% shorter. The length gain is real for a deployment that pays per token, but it is not evidence that evolution works.
2. **The search-level advantage of guided LLM mutation does not carry over to held-out fidelity.** Random search gains more fidelity on held-out events (+0.026) than NSGA-II with guided LLM mutation (+0.018). The rule-only versions and the unguided control gain +0.027 to +0.035. Random search does no selection during the search, but its knee is still the best of 30 sets on the search events by the knee rule. So it is a best-of-30 choice. The held-out fidelity gain is therefore better explained by choosing among candidates on the search events than by any particular mutation operator. Only the GA's gain (+0.043) is larger, and it does not survive correction. This is the main caution for the conclusions in Section 6.
3. **Disagreement hardly moves.** With the prompt set held fixed, B0's three sampling runs range from 0.406 to 0.433 on the held-out events. Every search arm stays within 0.021 of B0, except the unguided control, whose knees lose 0.035 (down to 0.372). Its raw p-value is 0.32. The idea that unguided rewriting makes the schools sound more alike fits Section 2.3, but this experiment cannot prove it.
4. **Search-level rankings do not carry over.** On the search events random search has the highest knee fidelity (0.474) and the GA the lowest (0.452); on the held-out events the order of these two is reversed (0.432 against 0.448). With five search events, the knee choice is driven partly by noise, which limits what the search-level comparison in Section 5.1 can say about generalisation.

5.3 Second model (Claude Haiku 5.5) and the published baseline

Every arm was re-run on Claude Haiku 5.5 with five seeds (addendum B), together with the published EvoPrompt-GA baseline (addendum C) and, after its result was seen, a length-bounded

variant of it (addendum D). Table 5 gives the search-level results (`results/haiku/summary_table.md`).

Method	Hypervolume	Front size	Knee fidelity	Knee disagreement	Knee words
NSGA-II + LLM	0.0744 ± 0.0096	7.0 ± 3.6	0.283 ± 0.011	0.471 ± 0.045	323 ± 33
GA (weighted sum)	0.0721 ± 0.0136	7.2 ± 1.9	0.263 ± 0.016	0.510 ± 0.058	330 ± 30
Random search	0.0664 ± 0.0052	5.8 ± 2.2	0.299 ± 0.008	0.455 ± 0.027	345 ± 26
NSGA-II, rule only	0.0675 ± 0.0145	7.8 ± 3.1	0.276 ± 0.040	0.491 ± 0.058	344 ± 24
GA, rule only	0.0677 ± 0.0107	7.2 ± 3.6	0.267 ± 0.033	0.509 ± 0.064	343 ± 19
NSGA-II + unguided LLM	0.0606 ± 0.0146	5.2 ± 1.9	0.290 ± 0.013	0.457 ± 0.071	368 ± 44
EvoPrompt (GA)	0.0444 ± 0.0099	3.0 ± 0.7	0.254 ± 0.020	0.456 ± 0.068	392 ± 28
EvoPrompt (GA), length-bounded	0.0491 ± 0.0085	6.2 ± 0.8	0.262 ± 0.020	0.483 ± 0.037	412 ± 45

The absolute fidelity is much lower than on qwen (0.25 to 0.30 against 0.45 to 0.47 on the search events): Haiku uses fewer of the lexicon terms, so fidelity values are not comparable across models, only within one. NSGA-II with guided LLM mutation again has the highest mean hypervolume. Table 5b repeats the questions of addendum A and adds the comparisons with the baseline; with five seeds the exact one-sided sign-flip p-value can reach 1/32 = 0.031.

Question	Comparison	Seeds 1 to 5	Mean	Consistent	p (one-sided)
Q1 selection, rule mutation	GA rule − NSGA-II rule	−0.007, +0.009, −0.003, +0.009, −0.007	+0.000	no	0.500
Q2 LLM mutation, GA selection	GA − GA rule	+0.018, +0.013, +0.010, −0.008, −0.011	+0.004	no	0.250
(Q2 under NSGA-II)	NSGA-II + LLM − NSGA-II rule	+0.012, +0.016, +0.005, +0.008, −0.006	+0.007	no	0.094
Q3 guidance	NSGA-II + LLM − NSGA-II + unguided LLM	+0.027, +0.013, −0.004, +0.000, +0.032	+0.014	no	0.094
Q4 published baseline	NSGA-II + LLM − EvoPrompt (GA)	+0.033, +0.024, +0.023, +0.024, +0.046	+0.030	yes	0.031

Question	Comparison	Seeds 1 to 5	Mean	Consistent	p (one-sided)
Q5 length-bounded baseline	NSGA-II + LLM − EvoPrompt (GA), length-bounded	+0.029, +0.023, +0.018, +0.013, +0.043	+0.025	yes	0.031

What carries over from qwen, and what does not:

- **Selection still does not matter** (Q1, and NSGA-II with LLM mutation against the GA: 0.0744 against 0.0721, with mixed signs).
- **The LLM effects are weaker and no longer consistent.** Guided mutation beats rule mutation in four of five seeds under NSGA-II, and in three of five under the GA. It beats the unguided control in four of five seeds (Q3, p = 0.094). The direction matches qwen, but the decision rule of addendum A (all seeds must agree) is not met. The logs show one reason: Haiku writes longer rewrites. The 120-word limit rejected 28 of 60 guided edits in NSGA-II (47%) and 27 of 54 in the GA (50%), against 27% and 29% on qwen, and 6 of 48 edits (13%) for the unguided control. So on Haiku the guided runs used rule mutation for about half of their LLM edits.
- **The original EvoPrompt baseline fails here, so the comparison says little.** NSGA-II with LLM mutation beats EvoPrompt-GA in all five seeds (Q4, p = 0.031). The reason is simple. EvoPrompt's children had 668 to 1,132 words (the initial population's median was 420). Every child hit the length limit of the objective, and only 3 of 120 were feasible, because outputs for such long prompts were often cut off or could not be parsed. Its archive never improved on the starting population (hypervolume 0.044 in every generation). The paper's operator asks for "a better prompt" and says nothing about length or output format. Those are exactly the constraints of this problem.
- **The length-limited version works as an optimiser, but still loses.** With one extra sentence in the template (addendum D, written after the result above), EvoPrompt's children had 396 to 608 words (median 480). 60 of 120 were feasible, no answer was rejected, and the archive improved every generation (from 0.044 to 0.049). NSGA-II with LLM mutation still beats it in all five seeds (Q5, mean +0.025, p = 0.031). Its children stay longer than the starting sets, so length and format still cost it. A general operator told only about length does worse than one told which of the three objectives to improve.

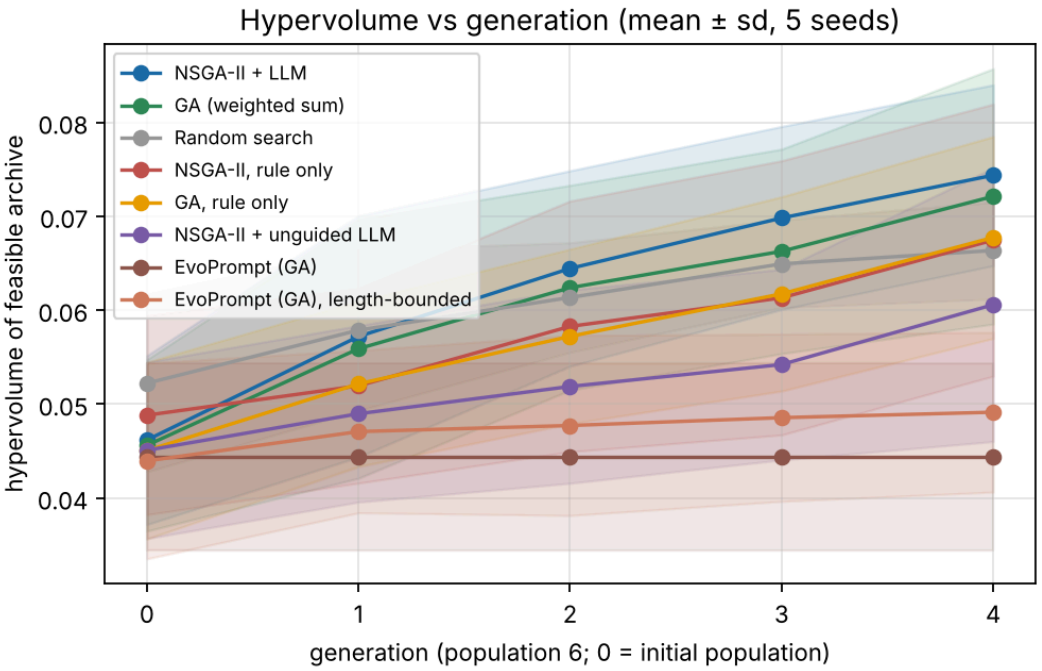


Figure 4. Mean hypervolume of the feasible archive per generation on Haiku (5 seeds, shaded ±1 sd). Initial populations differ slightly between arms because, after the cache was frozen, seeds 4 and 5 drew their initial samples separately per arm (Section 4.8).

On the held-out events (Table 6, `results/haiku/heldout_tests.csv`) no method differs from B0 after Holm correction over the method-versus-B0 tests; the smallest adjusted p-value is that of the unguided control's fidelity gain (+0.023, raw p = 0.008, adjusted 0.125 over 16 tests), and in the 72-test protocol family the smallest is 0.425. Neither EvoPrompt arm differs from NSGA-II + LLM on held-out events after correction. The same arm also raises disagreement (+0.058, raw p = 0.027), the opposite of its effect on qwen (−0.035), so the effect of unguided rewriting on disagreement is not stable across models. B0's five sampling replicates span 0.446 to 0.540 in held-out disagreement, wider than on qwen.

Table 6 gives the held-out means on Haiku (mean ± sd over five seeds; B0 over five sampling replicates).

Method	Fidelity	Disagreement	Validity	Words
B0 hand-written	0.238 ± 0.018	0.427 ± 0.067	0.927 ± 0.043	438
NSGA-II + LLM	0.248 ± 0.027	0.481 ± 0.039	0.960 ± 0.028	323 ± 33
GA (weighted sum)	0.244 ± 0.034	0.459 ± 0.031	0.940 ± 0.049	330 ± 30
Random search	0.259 ± 0.023	0.488 ± 0.058	0.933 ± 0.033	345 ± 26
NSGA-II, rule only	0.243 ± 0.020	0.458 ± 0.044	0.973 ± 0.028	344 ± 24
GA, rule only	0.241 ± 0.050	0.506 ± 0.046	0.967 ± 0.033	343 ± 19
NSGA-II + unguided LLM	0.261 ± 0.019	0.518 ± 0.069	0.967 ± 0.024	368 ± 44
EvoPrompt (GA)	0.224 ± 0.005	0.495 ± 0.040	0.947 ± 0.018	392 ± 28
EvoPrompt (GA), length-bounded	0.243 ± 0.010	0.463 ± 0.088	0.940 ± 0.028	412 ± 45

5.4 Qualitative observations

The examples below are from qwen. Summed over three seeds, 8 of 30 LLM-guided children in NSGA-II (27%) and 10 of 34 in the GA (29%) were rejected by the 5-to-120-word check and replaced by rule mutation (`results/runs/*/evaluations.jsonl`, origin suffix `f`). Reading the accepted edits (the fields of LLM children that do not occur earlier in the same run) shows two kinds of rewrite, which match the length and fidelity instructions.

- **Compression.** Length-directed edits shorten a diagnosis to about 40 words and mostly keep its structure. In the NSGA-II seed-3 knee, the Keynesian diagnosis went from 130 to 40 words: “Assess the output gap via unemployment (natural ~4.5%) and inflation (2% target): negative gaps need support; positive gaps require restraint. Distinguish supply shocks’ price versus income effects, noting which dominates. Evaluate policy using the real rate (policy rate minus inflation).” The definitions of the liquidity trap and the paradox of thrift, and the condition on a low policy rate, were dropped. Compression can also change the meaning. In the same run, the Monetarist sentence “a relative-price change becomes inflation only if money accommodates it” was shortened to “Money accommodates supply-shock inflation”, which reverses a conditional into a claim. The lexicon-based objectives cannot detect such an error.
- **Lexicon loading.** Fidelity-directed edits rewrite a diagnosis around school vocabulary and grow it to 80 to 110 words. A typical Austrian rewrite (NSGA-II, seed 3) reads “Ask first whether the event alters time preference or initiates credit expansion, for it is artificial credit expansion that distorts interest rates and launches the Austrian business cycle. Such distortion misguides entrepreneurs into lengthening the capital structure … generating widespread malinvestment.” These two sentences contain five of the six required Austrian lexicon terms (time preference, credit expansion, Austrian business cycle, capital structure, malinvestment); the B0 diagnosis it replaced contained two (credit expansion, malinvestment). This is the Goodhart risk named in Section 6.2 in its plain form: the edit raises the measured fidelity by naming the school’s concepts, and whether it improves the analysis is not measured.

Distinctness-directed edits are harder to recognise in the text. One GA rewrite of the Keynesian diagnosis (seed 2) defines the school against another one (“prioritizing demand management over concerns about capital misallocation”), which is the intended behaviour. The knee prompt sets of every run are in `results/best_prompts/`.

6 Discussion

6.1 What the results say

RQ2 (which part matters). On both models the selection method makes no difference at this budget, and NSGA-II with guided LLM mutation has the highest mean search hypervolume. On qwen the evidence for our operator is direct. Guided mutation beats rule mutation under both selection methods, and it beats LLM rewriting with no goal in every seed. The unguided control does no better than rule mutation. So the gain comes from the guidance, not from letting an LLM rewrite the text. On Haiku the same comparisons point the same way in four of five seeds, but they do not meet the decision rule fixed in addendum A. Half of the guided edits there were rejected by a length

limit tuned on qwen. The operator helps on the model it was designed with, and it helps less, but does not reverse, on a model that writes longer answers.

RQ4 (published baseline, second model). The original EvoPrompt-GA operator fails in this problem: it makes every prompt longer until the agents' outputs break. With one extra sentence limiting length (addendum D, decided after that result), the baseline works as an optimiser. NSGA-II with guided LLM mutation still reaches a higher hypervolume in all five seeds ($p = 0.031$). On held-out events, neither EvoPrompt arm differs from it after correction. The general lesson is that operators built for one prompt and one score do not carry over to a set of prompts that must meet a length objective and an output-format constraint. That is the gap identified in Section 2.

RQ1 and RQ3 (beating hand-written prompts, unseen events). On held-out events, no method beats the hand-written set on either model, and no held-out difference survives correction. The searched sets are 20% to 30% shorter, with similar fidelity and disagreement. But the shortening comes mostly from the operators and the knee rule (Section 5.2). The fidelity gains that appear before correction are as large for random search as for evolution. This points to the effect of choosing among candidates on the search events, not to any one operator.

6.2 Limitations

- **Budget.** Population 6 and four generations give 30 evaluations per run. This is a small study, reduced from the planned population 8 and five generations. In total the qwen runs made 3,709 calls (estimated USD 27.31 at assumed prices, including the smoke test and the incomplete EvoPrompt run) and the Haiku runs made 9,412 calls (USD 5.28 at list prices).
- **Seeds.** Three seeds per arm on qwen (smallest one-sided p 1/8) and five on Haiku (1/32). The search-level qwen findings are descriptive.
- **Models and sampling.** `qwen3.8-max` and Claude Haiku 5.5 are both used through an API, with different sampling (qwen 0.7 / 0.9, Haiku the provider default), so their absolute values are not comparable; only directions are compared. Outputs are cached but not bit-reproducible; the raw responses are committed so that every analysis can be repeated without the API.
- **Acceptance rule tuned to one model.** The 120-word limit on LLM edits rejected about 28% of guided edits on qwen and about half on Haiku. A limit expressed relative to the current text, or an instruction that states the limit, would remove this asymmetry.
- **Cost figures.** qwen ran on a prepaid Token Plan subscription, so its USD figures are estimates at assumed list prices, not charges; the Haiku figures use Anthropic's list prices. Token counts are measured for both.
- **Lexicon.** The fidelity objective uses a lexicon the thesis marks as PROVISIONAL. Optimising it invites Goodhart-style gaming (Section 5.4 shows it). Expert or model-based judging of faithfulness is the natural fix and is not included.
- **Synthetic events.** The events are generated, and their macro context is synthetic. The results say nothing about real forecasting accuracy.
- **Confounds.** LLM arms make more model calls per child than rule mutation, and EvoPrompt makes three per child. The comparison holds the budget fixed in evaluations, not in calls.
- **Single school family.** The three schools are one economic taxonomy. Generalisation to other perspectives is not tested.

- **Knee rule and operator bias.** The knee is chosen by unweighted distance, in which length carries more weight than fidelity, and rule mutation can only shorten the text. Both favour shorter prompt sets independently of search quality (Sections 3.3, 3.5 and 5.2).
- **Protocol history.** The main protocol was committed during the main run; the addenda were committed before their runs, but addendum C was amended after its first generation and addendum D responds to an observed result (Section 4.8). The qwen EvoPrompt arm is incomplete.

6.3 Future work

A run with more generations would test whether the selection scheme matters once trajectories have time to diverge, and a larger search set would give the held-out tests more power. A length rule relative to the current text would make the guided operator behave the same on models that write longer answers. Replacing the lexicon with a judge model calibrated on human ratings would address the Goodhart risk. A control that holds the number of model calls fixed would separate the LLM effect from the extra text it generates.

7 Conclusion

We framed the design of three competing economic-school agents as a constrained multi-objective search over sets of prompts. The objectives are fidelity to the school, disagreement between the schools, and prompt length. We proposed an LLM mutation that is told which goal to improve. In the main experiment the selection method did not matter at this budget. The goal-directed guidance did: guided mutation beat both rule mutation and an unguided LLM control in every seed. A second model with five seeds points the same way in four of five seeds, but less clearly, partly because a length limit tuned on the first model rejected half of its edits. The original EvoPrompt-GA method fails in this constrained problem. A length-limited version works, but still loses to our method in all five seeds on the second model. On unseen events, no method beats careful hand-written prompts after correction, and random search gains about as much fidelity as evolution does. The evidence supports goal-directed LLM mutation as the useful ingredient in prompt-set search. Two caveats apply: some fidelity edits game the lexicon, and some compressions change the meaning. The evidence does not show that evolved prompt sets beat careful hand-written ones on unseen events.

Appendix A: Use of AI tools

This appendix follows the course and PolyU guidance on disclosing generative AI; `AI_USE.md` in the submission archive gives the dated log.

- **Literature review.** AI sub-agents, one per theme, searched for 2024–2026 papers and opened the arXiv or venue page of each candidate. Author lists and first-submission dates were checked against arXiv on 2026-11-01, and journal DOIs, volumes and pages against Crossref on 2026-11-02. Venue details come from arXiv comments, venue pages or the course reference list.

- **Code.** The evo_prompt package, tests and runner were written with an AI coding assistant. They pass 27 offline tests (`python -m pytest -q courses/dsai6707/code`) and the repository's own test suite, and an automatic checker (`python -m evo_prompt.check_report`) compares the numbers in this report with the result files.
- **Analysis and figures.** Written with AI assistance; numbers are generated by the scripts in `code/evo_prompt/` and can be regenerated from `results/`.
- **Report prose.** Drafted with AI assistance. The author is responsible for the final wording, the interpretation and the citations.
- **Experiments.** The model calls were made by the code in this repository, through the two APIs, under the budget caps described in Section 4.2 and the protocol addenda. The AI assistant proposed the additional arms, the second model and the EvoPrompt baseline during a review of an earlier draft; each was fixed in an addendum before its runs (Section 4.8).

Appendix B: File map for the numbers

Number	File
One row per run (search and held-out metrics)	`results/summary.csv`
Hypervolume, front size, knee search metrics	`results/summary_search.csv`, `results/summary_table.md`
Held-out means, per-event values	`results/summary_table.md`, `results/heldout_per_event.csv`
Paired tests and Holm	`results/heldout_tests.csv` (method versus B0), `results/heldout_tests_protocol.csv` (protocol family)
Questions Q1 to Q5 (per-seed hypervolume differences)	`results/addendum_a_search.csv`, `results/haiku/addendum_a_search.csv`
Haiku replication (all tables and figures)	`results/haiku/` (same file names)
Hypervolume curves	`results/runs/<method>_s<seed>/gen_log.csv`, `results/figures/hv_vs_generation.png`
Evaluations and origins (LLM children, rejections)	`results/runs/<method>_s<seed>/evaluations.jsonl`
Best prompt sets	`results/best_prompts/<method>_s<seed>_knee.txt`
Raw model outputs	`results/cache/llm_cache.jsonl` (qwen); `results/haiku/cache/llm_cache.jsonl` and `results/haiku/cache/layers/` (Haiku)
Spend and token totals	`results/logs/*.log` and `results/haiku/logs/*.log` (ledger lines); recomputable from the caches
Smoke test	`results/smoke/nsga2_s1/`
Analysis protocol and addenda	`code/PROTOCOL.md`, `code/PROTOCOL_ADDENDUM_A.md` to `_D.md`

References

Akiba, Takuya, Makoto Shing, Yujin Tang, Qi Sun, and David Ha. 2025. “Evolutionary Optimization of Model Merging Recipes.” *Nature Machine Intelligence* 7: 195–204. https://doi.org/10.1038/s42256-024-00975-8.

Baltaji, Razan, Babak Hemmatian, and Lav R. Varshney. 2024. “Persona Inconstancy in Multi-Agent LLM Collaboration: Conformity, Confabulation, and Impersonation.” In *Proceedings of the 2nd Workshop on Cross-Cultural Considerations in NLP (C3NLP)*. https://arxiv.org/abs/2405.03862.

Câmara, Sara, Eduardo Luz, Valéria Carvalho, Ivan Meneghini, and Gladston Moreira. 2025. “MOPrompt: Multi-Objective Semantic Evolution for Prompt Optimization.” https://arxiv.org/abs/2508.01541.

Chen, Shilian, Jie Zhou, Tianyu Huai, Yujiang Lu, Junsong Li, Bihao Zhan, Qianjun Pan, et al. 2025. “Black-Box Model Merging for Language-Model-as-a-Service with Massive Model Repositories.” https://arxiv.org/abs/2509.12951.

Deb, Kalyanmoy, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. 2002. “A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II.” *IEEE Transactions on Evolutionary Computation* 6 (2): 182–97. https://doi.org/10.1109/4235.996017.

Fernando, Chrisantha, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. 2024. “Promptbreeder: Self-Referential Self-Improvement via Prompt Evolution.” In *Proceedings of the 41st International Conference on Machine Learning*, 235:13481–544. PMLR. https://arxiv.org/abs/2309.16797.

Forniés-Tabuenca, Diego, Alejandro Uribe, Urtzi Otamendi, Arkaitz Artetxe, Juan Carlos Rivera, and Oier Lopez de Lacalle. 2025. “REMoH: A Reflective Evolution of Multi-Objective Heuristics Approach via Large Language Models.” https://arxiv.org/abs/2506.07759.

Guo, Qingyan, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. 2024. “EvoPrompt: Connecting LLMs with Evolutionary Algorithms Yields Powerful Prompt Optimizers.” In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2309.08532.

Hu, Shengran, Cong Lu, and Jeff Clune. 2025. “Automated Design of Agentic Systems.” In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2408.08435.

Hu, Yuntong, Yuting Zhang, Matthew Trager, Yi Zhang, Shuo Yang, Wei Xia, and Stefano Soatto. 2026. “EvoMAS: Evolutionary Generation of Multi-Agent Systems.” In *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2602.06511.

Jiang, Liwei, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. 2025. “Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond).” In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*. https://arxiv.org/abs/2510.22954.

Kuroki, So, Taishi Nakamura, Takuya Akiba, and Yujin Tang. 2025. “Agent Skill Acquisition for Large Language Models via CycleQD.” In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2410.14735.

Lange, Robert Tjarko, Yingtao Tian, and Yujin Tang. 2024. “Large Language Models as Evolution Strategies.” https://arxiv.org/abs/2402.18381.

Lee, Donggyu, Hyeok Yun, Jungwon Kim, Junsik Min, Sungwon Park, Sangyoon Park, and Jihee Kim. 2026. “Ideological Bias in LLMs’ Economic Causal Reasoning.” In *Conference on*

Language Modeling (COLM). https://arxiv.org/abs/2604.21334.

Li, Shuyue Stella, Rui Xin, Teng Xiao, Yike Wang, Rulin Shao, Zoey Hao, Melanie Sclar, et al. 2026. “EvoLM: Self-Evolving Language Models Through Co-Evolved Discriminative Rubrics.” In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2605.03871.

Lim, Bryan, Manon Flageat, and Antoine Cully. 2024. “Large Language Models as in-Context AI Generators for Quality-Diversity.” https://arxiv.org/abs/2404.15794.

Lin, Jianhao, Lexuan Sun, and Yixin Yan. 2025. “Simulating Macroeconomic Expectations in Survey Experiments with LLM-Based Economic Agents.” https://arxiv.org/abs/2505.17648.

Liu, Fei, Xialiang Tong, Mingxuan Yuan, Xi Lin, Fu Luo, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. 2024. “Evolution of Heuristics: Towards Efficient Automatic Algorithm Design Using Large Language Model.” https://arxiv.org/abs/2401.02051.

Liu, Fei, Yiming Yao, Ping Guo, Zhiyuan Yang, Xi Lin, Zhe Zhao, Xialiang Tong, et al. 2026. “A Systematic Survey on Large Language Models for Algorithm Design.” *ACM Computing Surveys* 58 (8): 1–32. https://doi.org/10.1145/3787585.

Liu, Wanyi, Long Chen, and Zhenzhou Tang. 2024. “Large Language Model Aided Multi-Objective Evolutionary Algorithm: A Low-Cost Adaptive Approach.” https://arxiv.org/abs/2410.02301.

Ma, Yecheng Jason, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. “Eureka: Human-Level Reward Design via Coding Large Language Models.” In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2310.12931.

Qiu, Haiyun, Xingyu Wu, Liang Feng, and Kay Chen Tan. 2026. “Fine-Grained Model Merging via Modular Expert Recombination.” https://arxiv.org/abs/2602.06552.

Qiu, Xin, Yulu Gan, Conor F. Hayes, Qiyao Liang, Yinggan Xu, Roberto Dailey, Elliot Meyerson, Babak Hodjat, and Risto Miikkulainen. 2026. “Evolution Strategies at Scale: LLM Fine-Tuning Beyond Reinforcement Learning.” In *International Conference on Machine Learning (ICML)*. https://arxiv.org/abs/2509.24372.

Santos, Gabriel Machado, Rita Maria da Silva Julia, and Marcelo Zanchetta do Nascimento. 2025. “Diverse Prompts: Illuminating the Prompt Space of Large Language Models with MAP-Elites.” https://arxiv.org/abs/2504.14367.

Stein, Niki van, and Thomas Bäck. 2024. “LLaMEA: A Large Language Model Evolutionary Algorithm for Automatically Generating Metaheuristics.” https://arxiv.org/abs/2405.20132.

Takano, Kaito, Masanori Hirano, and Kei Nakagawa. 2025. “Modeling Hawkish-Dovish Latent Beliefs in Multi-Agent Debate-Based LLMs for Monetary Policy Decision Classification.” In *PRIMA 2025*. https://doi.org/10.1007/978-3-032-13562-9_38.

Wang, Yuhan, Chaoda Peng, Xingyu Wu, Sheng-Hao Wu, and Zhi-Hui Zhan. 2026. “How to Guide LLM Generation: Dual-Surrogate Guided Search for Automated Heuristic Design.” https://arxiv.org/abs/2607.13911.

Wu, Haolun, Zhenkun Li, and Lingyao Li. 2025. “Can LLM Agents Really Debate? A Controlled Study of Multi-Agent Debate in Logical Reasoning.” https://arxiv.org/abs/2511.07784.

Wu, Xingyu, Sheng-hao Wu, Jibin Wu, Liang Feng, and Kay Chen Tan. 2025. “Evolutionary Computation in the Era of Large Language Model: Survey and Roadmap.” *IEEE Transactions on Evolutionary Computation* 29 (2): 534–54. https://doi.org/10.1109/TEVC.2024.3506731.

Wu, Xingyu, Yu Zhou, and KC Tan. 2026. “Building LLMs Like LEGO: Two-Dimensional Architecture Reassembly of Large Language Models.” In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 44939–56. https://doi.org/10.18653/v1/2026.acl-long.2081.

Ye, Haoran, Jiarui Wang, Zhiguang Cao, Federico Berto, Chuanbo Hua, Haeyeon Kim, Jinkyoo Park, and Guojie Song. 2024. “ReEvo: Large Language Models as Hyper-Heuristics with Reflective Evolution.” In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2402.01145.

Zhang, Yiming, Baoyi He, Shengyu Zhang, Yuhao Fu, Qi Zhou, Zhijie Sang, Zijin Hong, et al. 2024. “Unconstrained Model Merging for Enhanced LLM Reasoning.” https://arxiv.org/abs/2410.13699.

Zhao, Guang, Byung-Jun Yoon, Gilchan Park, Shantenu Jha, Shinjae Yoo, and Xiaoning Qian. 2025. “Pareto Prompt Optimization.” In *International Conference on Learning Representations (ICLR)*. https://proceedings.iclr.cc/paper_files/paper/2025/hash/13b45b44e26c353c64cba9529bf4724f-Abstract-Conference.html.

Zhou, Han, Xingchen Wan, Ruoxi Sun, Hamid Palangi, Shariq Iqbal, Ivan Vulić, Anna Korhonen, and Sercan Ö. Arık. 2026. “Multi-Agent Design: Optimizing Agents with Better Prompts and Topologies.” In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/2502.02533.

Zhu, Xiaochen, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier, and Andreas Vlachos. 2026. “Demystifying Multi-Agent Debate: The Role of Confidence and Diversity.” https://arxiv.org/abs/2601.19921.